

“C PROGRAMMING”

ASSIGNMENT – 1

OPERATORS

1. Arithmetic Operators + - * / %

- I. Write a C program to input two numbers and print their sum, difference, product, quotient, and remainder.

// Program to input two numbers and print their sum, difference, product, quotient, and remainder

```
#include <stdio.h>
int main()
{
    int a, b, sum, difference, product, quotient, remainder;

    printf("Enter the first number: ");
    scanf("%d", &a);

    printf("Enter the second number: ");
    scanf("%d", &b);

    sum=a+b;
    printf("Sum= %d\n", sum);

    difference=a-b;
    printf("Difference= %d\n", difference);

    product=a*b;
    printf("Product= %d\n", product);

    quotient=a/b;
    printf("Quotient= %d\n", quotient);

    remainder=a%b;
    printf("Remainder= %d\n", remainder);
```

```
    return 0;  
}
```

Output :

```
Enter the first number: 34  
Enter the second number: 32  
Sum= 66  
Difference= 2  
Product= 1088  
Quotient= 1  
Remainder= 2  
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

II. Write a program to calculate the area and perimeter of a rectangle.

```
// Program to calculate the area and perimeter of a rectangle

#include <stdio.h>

int main()
{
    int l, b, a, p;

    printf("Enter the length of the rectangle: ");

    scanf("%d", &l);

    printf("Enter the breadth of the rectangle: ");

    scanf("%d", &b);

    a=l*b;

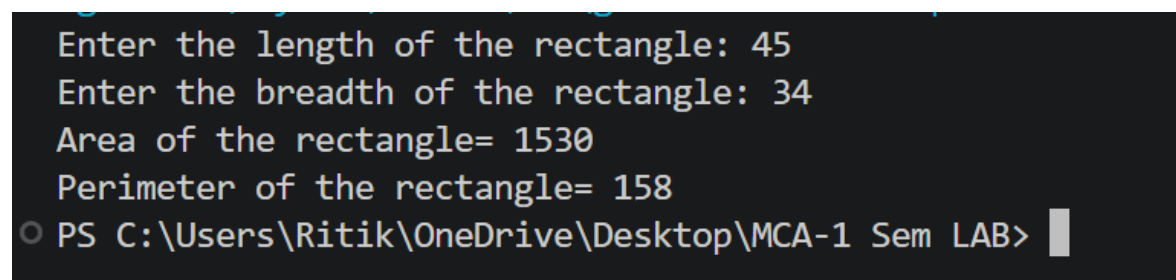
    printf("Area of the rectangle= %d\n", a);

    p=2*(l+b);

    printf("Perimeter of the rectangle= %d\n", p);

    return 0;
}
```

Output :



```
Enter the length of the rectangle: 45
Enter the breadth of the rectangle: 34
Area of the rectangle= 1530
Perimeter of the rectangle= 158
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

III. Write a program to calculate the simple interest using arithmetic operators.

```
// Program to calculate the simple interest using arithmetic operators
```

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int p, t;
```

```
    float r, si;
```

```
    printf("Enter the principal: ");
```

```
    scanf("%d", &p);
```

```
    printf("Enter the rate: ");
```

```
    scanf("%f", &r);
```

```
    printf("Enter the time: ");
```

```
    scanf("%d", &t);
```

```
    si= (p*r*t)/100;
```

```
    printf("Simple Interest= %.2f\n",si);
```

```
    return 0;
```

```
}
```

Output

```
Enter the principal: 32
```

```
Enter the rate: 2.5
```

```
Enter the time: 4
```

```
Simple Interest= 3.20
```

```
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

IV. Write a program to input a 3-digit number and find the sum of its digits.

// Program to input a 3-digit number and find the sum of its digits

```
#include <stdio.h>

int main()
{
    int dig, sum;

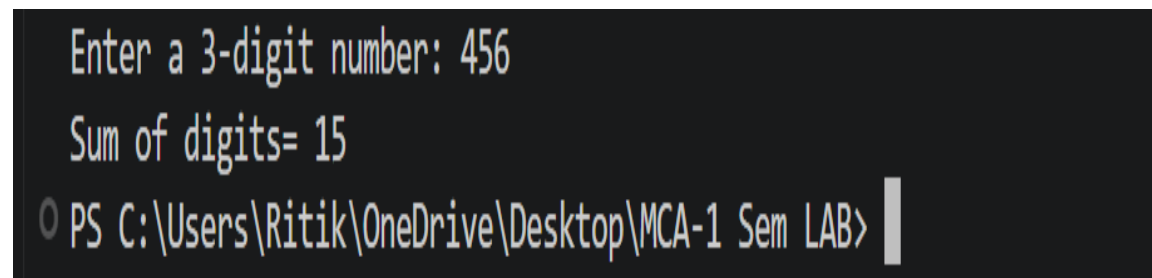
    printf("Enter a 3-digit number: ");
    scanf("%d", &dig);

    sum=(dig/100)+((dig/10)%10)+(dig%10);

    printf("Sum of digits= %d\n", sum);

    return 0;
}
```

Output:

A screenshot of a terminal window with a dark background. The text is displayed in a light green monospaced font. The first line shows the prompt 'Enter a 3-digit number:' followed by the input '456'. The second line shows the output 'Sum of digits= 15'. The third line shows the command prompt 'PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>' followed by a vertical bar cursor.

```
Enter a 3-digit number: 456
Sum of digits= 15
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB> |
```

V. Write a program to convert seconds into hours, minutes, and seconds.

// Program to convert seconds into hours, minutes and seconds

```
#include <stdio.h>

int main()
{
    int seconds, hours, mins, secs;

    printf("Enter the seconds: ");
    scanf("%d", &seconds);

    hours=seconds/3600;
    printf("Hours= %d\n", hours);

    mins=(seconds%3600)/60;
    printf("Minutes= %d\n", mins);

    secs=seconds%60;
    printf("Seconds= %d\n", secs);

    return 0;
}
```

Output:

```
Enter the seconds: 7854
Hours= 2
Minutes= 10
Seconds= 54
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB> □
```

2. Relational Operators

- VI. Write a program to input two numbers and check which number is greater.

// Program to input two numbers and check which number is greater

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int n1, n2;
```

```
    printf("Enter the first number: ");
```

```
    scanf("%d", &n1);
```

```
    printf("Enter the second number: ");
```

```
    scanf("%d", &n2);
```

```
    if(n1>n2)
```

```
    {
```

```
        printf("Greater Number= %d\n", n1);
```

```
    }
```

```
    else if(n1<n2)
```

```
    {
```

```
        printf("Greater Number= %d\n", n2);
```

```
    }
```

```
    else
```

```
    {
```

```
        printf("Both numbers are equal= %d\n", n1);
```

```
    }
```

```
    return 0;
```

}

Output:

```
Enter the first number: 23
```

```
Enter the second number: 34
```

```
Greater Number= 34
```

```
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

VII. Write a program to check whether a number is equal to zero, positive, or negative.

// Program to check whether a number is equal to zero, positive, or negative

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int n;
```

```
    printf("Enter the number: ");
```

```
    scanf("%d", &n);
```

```
    if(n==0)
```

```
    {
```

```
        printf("Number is equal to zero\n");
```

```
    }
```

```
    else if(n>0)
```

```
    {
```

```
        printf("Number is positive\n");
```

```
    }
```

```
    else
```

```
    {
```

```
        printf("Number is negative\n");
```

```
    }
```

```
    return 0;
```

```
}
```

Output :

```
● bgExe=C:\msys64\ucrt64\bin\gdb.exe --interpreter=mi
Enter the number: 4
Number is positive
● PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB> ${env:DEBUGINFOD_URLS}=''; & 'c:\Users\Ritik\
.vscode\extensions\ms-vscode.cpptools-1.34.4-win32-x64\debugAdapters\bin\WindowsDebugLauncher.ex
e' '--stdin=Microsoft-MIEngine-In-cpsf3nqk.tdd' '--stdout=Microsoft-MIEngine-Out-0v5aavfi.i4r' '
--stderr=Microsoft-MIEngine-Error-xdrap1nc.4he' '--pid=Microsoft-MIEngine-Pid-uvmlxvsy.wo5' '--d
bgExe=C:\msys64\ucrt64\bin\gdb.exe' '--interpreter=mi'
Enter the number: 0
Number is equal to zero
● PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB> ${env:DEBUGINFOD_URLS}=''; & 'c:\Users\Ritik\
.vscode\extensions\ms-vscode.cpptools-1.34.4-win32-x64\debugAdapters\bin\WindowsDebugLauncher.ex
e' '--stdin=Microsoft-MIEngine-In-51ug0pyz.ukf' '--stdout=Microsoft-MIEngine-Out-1jy2rwws.05t' '
--stderr=Microsoft-MIEngine-Error-a52e4akq.xs3' '--pid=Microsoft-MIEngine-Pid-24zzgtjt.wfo' '--d
bgExe=C:\msys64\ucrt64\bin\gdb.exe' '--interpreter=mi'
Enter the number: -9
Number is negative
● PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

VIII. Write a program to input a student's marks and check whether the student has passed or failed.

// Program to input a student's marks and check whether the student has passed or failed

```
#include <stdio.h>
int main()
{
    int marks;
    printf("Enter the marks of the student: ");
    scanf("%d", &marks);

    if(marks >= 45)
    {
        printf("Student is passed\n");
    }
    else{
        printf("Student is failed\n");
    }
    return 0;
}
```

Output:

```
Enter the marks of the student: 67
```

```
Student is passed
```

```
● PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB> ${env:DEBUGINFOD_URLS}=''; & 'c:\Users\Ritik\OneDrive\Desktop\vscode\extensions\ms-vscode.cpptools-1.34.4-win32-x64\debugAdapters\bin\WindowsDebugLauncher.exe' '--stdin=Microsoft-MIEngine-In-2xddgugm.uhm' '--stdout=Microsoft-MIEngine-Out-n0uy2f5f.yid' '--stderr=Microsoft-MIEngine-Error-gow0fp5c.cg0' '--pid=Microsoft-MIEngine-Pid-fiwi2yqv.0lj' '--dbgExe=C:\msys64\ucrt64\bin\gdb.exe' '--interpreter=mi'
```

```
Enter the marks of the student: 34
```

```
Student is failed
```

```
○ PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB> █
```

IX. Write a program to find the largest of two numbers using relational operators.

// Program to find the largest of two numbers using relational operators

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int n1, n2;
```

```
    printf("Enter the first number: ");
```

```
    scanf("%d", &n1);
```

```
    printf("Enter the second number: ");
```

```
    scanf("%d", &n2);
```

```
    if(n1 > n2)
```

```
    {
```

```
        printf("Largest Number = %d\n", n1);
```

```
    }
```

```
    else if(n2 > n1)
```

```
    {
```

```
        printf("Largest Number = %d\n", n2);
```

```
    }
```

```
    else
```

```
    {
```

```
        printf("Both numbers are equal = %d\n", n1);
```

```
    }
```

```
    return 0;
```

```
}
```

Output:

```
Enter the first number: 23
Enter the second number: 55
Largest Number = 55
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB> |
```

X. Write a program to check whether a given year is a leap year.

// Program to check whether a given year is a leap year

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int year;
```

```
    printf("Enter the year: ");
```

```
    scanf("%d", &year);
```

```
    if(year % 400 == 0)
```

```
    {
```

```
        printf("This year is a leap year");
```

```
    }
```

```
    else if(year % 100 == 0)
```

```
    {
```

```
        printf("This year is not a leap year");
```

```
    }
```

```
    else if(year % 4 == 0)
```

```
    {
```

```
        printf("This year is a leap year");
```

```
    }
```

```
    else
```

```
    {
```

```
        printf("This year is not a leap year");
```

```
    }
```

```
    return 0;
```

```
}
```

Output :

Enter the year: 2004

This year is a leap year

```
● PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB> ${env:DEBUGINFOD_URLS}=''; & 'c:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB\vscode\extensions\ms-vscode.cpptools-1.34.4-win32-x64\debugAdapters\bin\WindowsDebugLauncher.exe' '--stdin=Microsoft-MIEngine-In-3vmekooz.fbu' '--stdout=Microsoft-MIEngine-Out-or3ywxkp.lks' '--stderr=Microsoft-MIEngine-Error-frw0kvr3.xjs' '--pid=Microsoft-MIEngine-Pid-bxmpqlnh.3mm' '--dbgExe=C:\msys64\ucrt64\bin\gdb.exe' '--interpreter=mi'
```

Enter the year: 2026

This year is not a leap year

```
○ PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB> 
```

3. Logical Operators

XI. Write a program to check whether a number lies between 10 and 50.

// Program to check whether a number lies between 10 and 50

```
#include <stdio.h>
int main()
{
    int n;
    printf("Enter the number to check: ");
    scanf("%d", &n);

    if(n>=10 && n<=50)
    {
        printf("This number lies between 10 and 50\n");
    }
    else{
        printf("This number doesn't lie between 10 and 50\n");
    }
    return 0;
}
```

Output:

Enter the number to check: 35

This number lies between 10 and 50

```
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB> ${env:DEBUGINFOD_URLS}=''; & 'c:\Users\Ritik\
.vscode\extensions\ms-vscode.cpptools-1.34.4-win32-x64\debugAdapters\bin\WindowsDebugLauncher.ex
e' '--stdin=Microsoft-MIEngine-In-yyz0ocpc.exn' '--stdout=Microsoft-MIEngine-Out-o2sr0tnb.vgd' '
--stderr=Microsoft-MIEngine-Error-xhp4bx35.5hv' '--pid=Microsoft-MIEngine-Pid-cw5rxutz.kll' '--d
bgExe=C:\msys64\ucrt64\bin\gdb.exe' '--interpreter=mi'
```

Enter the number to check: 3

This number doesn't lie between 10 and 50

```
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB> █
```

XII. Write a program to check whether a person is eligible to vote based on age.

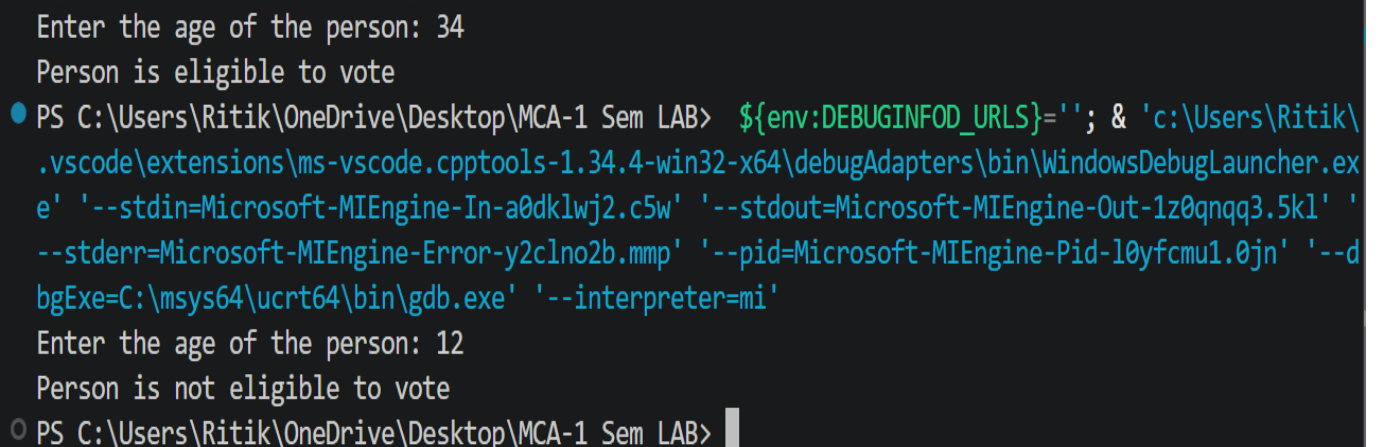
// Program to check whether a person is eligible to vote based on age

```
#include <stdio.h>
int main()
{
    int age;
    printf("Enter the age of the person: ");
    scanf("%d", &age);

    if(age>=18)
    {
        printf("Person is eligible to vote\n");
    }
    else
    {
        printf("Person is not eligible to vote\n");
    }
    return 0;}

```

Output:



```
Enter the age of the person: 34
Person is eligible to vote
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB> ${env:DEBUGINFOD_URLS}=''; & 'c:\Users\Ritik\
.vscode\extensions\ms-vscode.cpptools-1.34.4-win32-x64\debugAdapters\bin\WindowsDebugLauncher.ex
e' '--stdin=Microsoft-MIEngine-In-a0dklwj2.c5w' '--stdout=Microsoft-MIEngine-Out-1z0qnqq3.5kl' '
--stderr=Microsoft-MIEngine-Error-y2c1no2b.mmp' '--pid=Microsoft-MIEngine-Pid-10yfcmu1.0jn' '--d
bgExe=C:\msys64\ucrt64\bin\gdb.exe' '--interpreter=mi'
Enter the age of the person: 12
Person is not eligible to vote
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>

```

XIII. Write a program to check whether a number is divisible by both 3 and 5.

// Program to check whether a number is divisible by both 3 and 5

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int n;
```

```
    printf("Enter the number to check: ");
```

```
    scanf("%d", &n);
```

```
    if(n%3==0 && n%5==0)
```

```
    {
```

```
        printf("Number is divisible by both 3 and 5\n");
```

```
    }
```

```
    else
```

```
    {
```

```
        printf("Number is not divisible by both 3 and 5\n");
```

```
    }
```

```
    return 0;
```

Output:

```
Enter the number to check: 15
```

```
Number is divisible by both 3 and 5
```

```
● PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB> ${env:DEBUGINFOD_URLS}=''; & 'c:\Users\Ritik\OneDrive\Desktop\Code\vscode\extensions\ms-vscode.cpptools-1.34.4-win32-x64\debugAdapters\bin\WindowsDebugLauncher.exe' '--stdin=Microsoft-MIEngine-In-f1wg4oxm.dzq' '--stdout=Microsoft-MIEngine-Out-larj3npv.5e4' '--stderr=Microsoft-MIEngine-Error-s30usv5d.rt2' '--pid=Microsoft-MIEngine-Pid-l3wsdie5.lor' '--command=C:\msys64\ucrt64\bin\gdb.exe' '--interpreter=mi'
```

```
Enter the number to check: 9
```

```
Number is not divisible by both 3 and 5
```

```
○ PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB> █
```

XIV. Write a program to check whether a number is positive and even.

// Program to check whether a number is positive and even

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int n;
```

```
    printf("Enter the number to check: ");
```

```
    scanf("%d", &n);
```

```
    if(n>0 && n%2==0)
```

```
    {
```

```
        printf("Number is positive and even");
```

```
    }
```

```
    else
```

```
    {
```

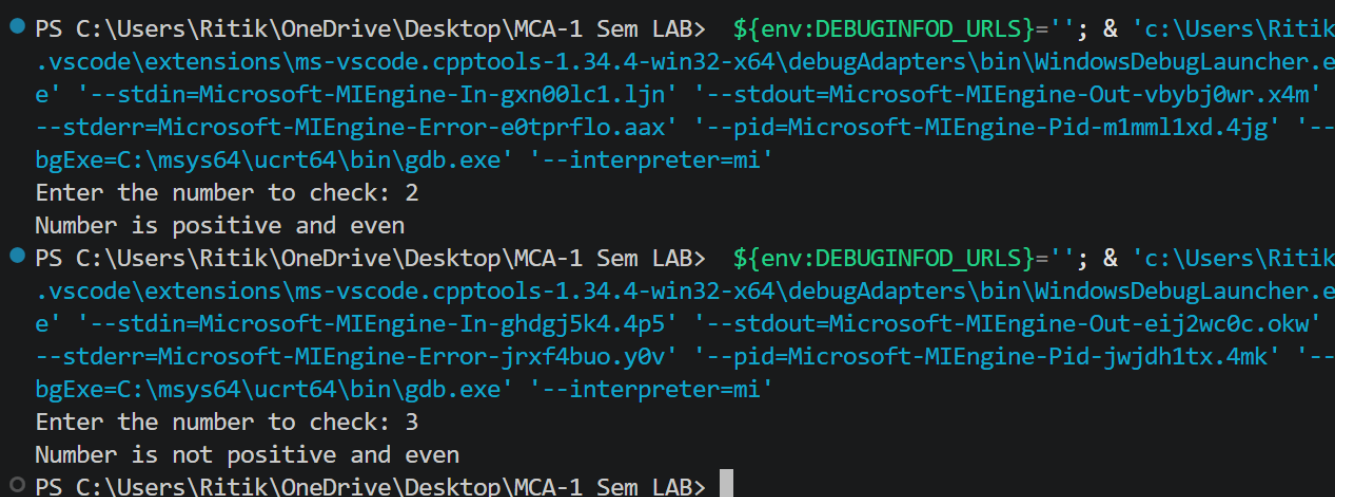
```
        printf("Number is not positive and even");
```

```
    }
```

```
    return 0;
```

```
}
```

Output:



```
● PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB> ${env:DEBUGINFOD_URLS}=''; & 'c:\Users\Ritik\OneDrive\Desktop\vscode\extensions\ms-vscode.cpptools-1.34.4-win32-x64\debugAdapters\bin\WindowsDebugLauncher.exe' '--stdin=Microsoft-MIEngine-In-gxn00lc1.ljn' '--stdout=Microsoft-MIEngine-Out-vbybj0wr.x4m' '--stderr=Microsoft-MIEngine-Error-e0tprflo.aax' '--pid=Microsoft-MIEngine-Pid-m1mm11xd.4jg' '--bgExe=C:\msys64\ucrt64\bin\gdb.exe' '--interpreter=mi'
Enter the number to check: 2
Number is positive and even
● PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB> ${env:DEBUGINFOD_URLS}=''; & 'c:\Users\Ritik\OneDrive\Desktop\vscode\extensions\ms-vscode.cpptools-1.34.4-win32-x64\debugAdapters\bin\WindowsDebugLauncher.exe' '--stdin=Microsoft-MIEngine-In-ghdgj5k4.4p5' '--stdout=Microsoft-MIEngine-Out-eij2wc0c.okw' '--stderr=Microsoft-MIEngine-Error-jrxf4buo.y0v' '--pid=Microsoft-MIEngine-Pid-jwjdh1tx.4mk' '--bgExe=C:\msys64\ucrt64\bin\gdb.exe' '--interpreter=mi'
Enter the number to check: 3
Number is not positive and even
○ PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB> █
```

XV. Write a program to check whether a character is a vowel or consonant using logical operators.

// Program to check whether a character is a vowel or consonant using logical operators

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    char c;
```

```
    printf("Enter the character to check: ");
```

```
    scanf(" %c", &c);
```

```
    if(c=='a' || c=='e' || c=='i' || c=='o' || c=='u' || c=='A' || c=='E' || c=='I' || c=='O' || c=='U')
```

```
    {
```

```
        printf("Character is vowel");
```

```
    }
```

```
    else
```

```
    {
```

```
        printf("Character is consonant");
```

```
    }
```

```
    return 0;
```

```
Enter the character to check: A
Character is vowel
● PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB> ${env:DEBUGINFOD_URLS}='' & 'c:\Users\Riti
.vscode\extensions\ms-vscode.cpptools-1.34.4-win32-x64\debugAdapters\bin\WindowsDebugLauncher.
e' '--stdin=Microsoft-MIEngine-In-pq1pra3p.y34' '--stdout=Microsoft-MIEngine-Out-iqnrnmzv.ynu'
--stderr=Microsoft-MIEngine-Error-x3grylgi.dig' '--pid=Microsoft-MIEngine-Pid-i5se0me1.ksl' '-
bgExe=C:\msys64\ucrt64\bin\gdb.exe' '--interpreter=mi'
Enter the character to check: a
Character is vowel
● PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB> ${env:DEBUGINFOD_URLS}='' & 'c:\Users\Riti
.vscode\extensions\ms-vscode.cpptools-1.34.4-win32-x64\debugAdapters\bin\WindowsDebugLauncher.
e' '--stdin=Microsoft-MIEngine-In-sjzt1pqp.0fb' '--stdout=Microsoft-MIEngine-Out-j4tbqso5.0zc'
--stderr=Microsoft-MIEngine-Error-5dhmlfvs.xiw' '--pid=Microsoft-MIEngine-Pid-wy0ydz4t4.by0' '-
bgExe=C:\msys64\ucrt64\bin\gdb.exe' '--interpreter=mi'
Enter the character to check: q
Character is consonant
○ PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB> }
```

4. Assignment Operators

XVI. Write a program to demonstrate all compound assignment operators.

// Program to demonstrate compound assignment operators

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int n;
```

```
    printf("Enter the number: ");
```

```
    scanf("%d", &n);
```

```
    n += 30;
```

```
    printf("After += : %d\n", n);
```

```
    n -= 25;
```

```
    printf("After -= : %d\n", n);
```

```
    n *= 20;
```

```
    printf("After *= : %d\n", n);
```

```
    n /= 15;
```

```
    printf("After /= : %d\n", n);
```

```
    n %= 10;
```

```
    printf("After %%= : %d\n", n);
```

```
    return 0;
```

```
}
```

Output:

```
Enter the number: 34
After += : 64
After -= : 39
After *= : 780
After /= : 52
After %= : 2
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

XVII. Write a program that increases a number by 10 using +=.

// Program that increases a number by 10 using +=

```
#include <stdio.h>

int main()
{
    int n;

    printf("Enter the number: ");
    scanf("%d", &n);

    n+= 10;

    printf("Number after using += = %d\n", n);

    return 0;
}
```

Output:

```
Enter the number: 8
Number after using += = 18
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

XVIII. Write a program that decreases a number by 5 using -=.

// Program that decreases a number by 5 using -=

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int n;
```

```
    printf("Enter the number: ");
```

```
    scanf("%d", &n);
```

```
    n-= 5;
```

```
    printf("Number after using -= = %d\n", n);
```

```
    return 0;
```

```
}
```

Output:

```
Enter the number: 10
Number after using -= = 5
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB> █
```

XIX. Write a program to calculate the final value of a variable after applying +=, -=, *=, and /=.

// Program to calculate the final value of a variable after applying +=, -=, *=, and /=

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int n;
```

```
    printf("Enter the number: ");
```

```
    scanf("%d", &n);
```

```
    n+= 20;
```

```
    n-= 15;
```

```
    n*= 10;
```

```
    n/= 5;
```

```
    printf("Final Value of a variable: %d\n", n);
```

```
    return 0;
```

```
}
```

Output:

```
Enter the number: 5
```

```
Final Value of a variable: 20
```

```
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

5. Bitwise Operators

XX. Write a C program to find the result of bitwise AND of two numbers.

// Program to find the result of bitwise AND of two numbers

```
#include <stdio.h>

int main()
{
    int n1, n2, result;

    printf("Enter the first number: ");
    scanf("%d", &n1);

    printf("Enter the second number: ");
    scanf("%d", &n2);

    result=n1&n2;

    printf("Result of bitwise AND of two numbers= %d\n", result);
    return 0;
}
```

Output:

```
Enter the first number: 2
Enter the second number: 3
Result of bitwise AND of two numbers= 2
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

XXI. Write a program to demonstrate AND, OR, and XOR operators.

// Program to demonstrate AND, OR, and XOR operators.

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int n1, n2, and, or, xor;
```

```
    printf("Enter the first number: ");
```

```
    scanf("%d", &n1);
```

```
    printf("Enter the second number: ");
```

```
    scanf("%d", &n2);
```

```
    and=n1&n2;
```

```
    printf("AND of %d and %d= %d\n", n1, n2, and);
```

```
    or=n1|n2;
```

```
    printf("OR of %d and %d= %d\n", n1, n2, or);
```

```
    xor=n1^n2;
```

```
    printf("XOR of %d and %d= %d\n", n1, n2, xor);
```

```
    return 0;
```

```
}
```

Output:

```
Enter the first number: 22
Enter the second number: 33
AND of 22 and 33= 0
OR of 22 and 33= 55
XOR of 22 and 33= 55
```

```
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

XXII. Write a program to check whether a number is even or odd using bitwise AND.

```
// Program to check whether a number is even or odd using bitwise AND
```

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int n;
```

```
    printf("Enter a number: ");
```

```
    scanf("%d", &n);
```

```
    if ((n & 1) == 0)
```

```
    {
```

```
        printf("%d is Even", n);
```

```
    }
```

```
    else
```

```
    {
```

```
        printf("%d is Odd", n);
```

```
    }
```

```
    return 0;
```

Output:

```
Enter a number: 2
2 is Even
● PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB> ${env:DEBUGINFOD_URLS}=''; & 'c:\Users\Ritik\
.vscode\extensions\ms-vscode.cpptools-1.34.4-win32-x64\debugAdapters\bin\WindowsDebugLauncher.ex
e' '--stdin=Microsoft-MIEngine-In-ncmovswx4' '--stdout=Microsoft-MIEngine-Out-kus0kgyl.11d' '
--stderr=Microsoft-MIEngine-Error-2frlh40n.03t' '--pid=Microsoft-MIEngine-Pid-dp1wz2q2.2pc' '--d
bgExe=C:\msys64\ucrt64\bin\gdb.exe' '--interpreter=mi'
Enter a number: 5
5 is Odd
○ PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB> █
```

XXIII. Write a program to multiply a number by 2 using the left shift operator.
// Program to multiply a number by 2 using the left shift operator

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int n, result;
```

```
    printf("Enter a number: ");
```

```
    scanf("%d", &n);
```

```
    result = n << 1;
```

```
    printf("Number after multiplying by 2 = %d", result);
```

```
    return 0;
```

```
}
```

Output:

```
Enter a number: 45
```

```
Number after multiplying by 2 = 90
```

```
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

XXIV. Write a program to divide a number by 2 using the right shift operator.

```
// Program to divide a number by 2 using the right shift operator
```

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int n, result;
```

```
    printf("Enter a number: ");
```

```
    scanf("%d", &n);
```

```
    result = n >> 1;
```

```
    printf("Number after dividing by 2 = %d", result);
```

```
    return 0;
```

```
}
```

Output:

```
Enter a number: 32
```

```
Number after dividing by 2 = 16
```

```
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

6. Conditional Operator

XXV. Write a program to find the largest of two numbers using the conditional operator.

// Program to find the largest of two numbers using the conditional operator

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int a, b, largest;
```

```
    printf("Enter first number: ");
```

```
    scanf("%d", &a);
```

```
    printf("Enter second number: ");
```

```
    scanf("%d", &b);
```

```
    largest = (a > b) ? a : b;
```

```
    printf("Largest number = %d", largest);
```

```
    return 0;
```

```
}
```

Output:

```
Enter first number: 54
Enter second number: 24
Largest number = 54
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

XXVI. Write a program to check whether a number is even or odd using ?:.

// Program to check whether a number is even or odd using ?: operator

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int n;
```

```
    printf("Enter a number: ");
```

```
    scanf("%d", &n);
```

```
    (n % 2 == 0) ? printf("%d is Even", n) : printf("%d is Odd", n);
```

```
    return 0;}
```

Output:

```
Enter a number: 3
```

```
3 is Odd
```

```
● PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB> ${env:PATH}
.vscod\extensions\ms-vscode.cpptools-1.34.4-win32-x64\de
e' '--stdin=Microsoft-MIEngine-In-wa2r0lkf.bgc' '--stdou
--stderr=Microsoft-MIEngine-Error-m02da2pp.b4i' '--pid=M
bgExe=C:\msys64\ucrt64\bin\gdb.exe' '--interpreter=mi'
```

```
Enter a number: 4
```

```
4 is Even
```

```
○ PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

KXVII. Write a program to find the smallest of two numbers using ?:.

// Program to find the smallest of two numbers using ?:

```
#include <stdio.h>

int main()
{
    int a, b, smallest;

    printf("Enter first number: ");
    scanf("%d", &a);

    printf("Enter second number: ");
    scanf("%d", &b);

    smallest = (a < b) ? a : b;

    printf("Smallest number = %d", smallest);
    return 0;
}
```

Output:

```
Enter first number: 34
Enter second number: 21
Smallest number = 21
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

XVIII. Write a program to check whether a person is eligible or not eligible to vote using ?:.

// Program to check whether a person is eligible or not eligible to vote using ?:

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int age;
```

```
    printf("Enter your age: ");
```

```
    scanf("%d", &age);
```

```
    (age >= 18) ? printf("Eligible to vote") : printf("Not eligible to vote");
```

```
    return 0;
```

```
}
```

Output:

```
Enter your age: 34
Eligible to vote
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

```
Enter your age: 10
Not eligible to vote
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

XXIX. Write a program to find the largest of three numbers using nested conditional operators.

```
// Program to find the largest of three numbers using nested conditional operators
```

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int a, b, c, largest;
```

```
    printf("Enter first number: ");
```

```
    scanf("%d", &a);
```

```
    printf("Enter second number: ");
```

```
    scanf("%d", &b);
```

```
    printf("Enter third number: ");
```

```
    scanf("%d", &c);
```

```
    largest = (a > b) ? ((a > c) ? a : c) : ((b > c) ? b : c);
```

```
    printf("Largest number = %d", largest);
```

```
    return 0;
```

```
}
```

Output:

```
Enter first number: 34
Enter second number: 65
Enter third number: 78
Largest number = 78
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB> |
```

7. Increment and Decrement Operators

XXX. Write a program to demonstrate pre-increment and post-increment.

```
// Program to demonstrate pre-increment and post-increment
```

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int a;
```

```
    printf("Enter a number: ");
```

```
    scanf("%d", &a);
```

```
    printf("Pre-increment = %d\n", ++a);
```

```
    printf("Post-increment = %d\n", a++);
```

```
    printf("Value after post-increment = %d", a);
```

```
    return 0;
```

```
}
```

Output:

```
Enter a number: 20
Pre-increment = 21
Post-increment = 21
Value after post-increment = 22
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

XXXI. Write a program to demonstrate pre-decrement and post-decrement.

// Program to demonstrate pre-decrement and post-decrement

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int a;
```

```
    printf("Enter a number: ");
```

```
    scanf("%d", &a);
```

```
    printf("Pre-decrement = %d\n", --a);
```

```
    printf("Post-decrement = %d\n", a--);
```

```
    printf("Value after post-decrement = %d", a);
```

```
    return 0;
```

```
}
```

Output:

```
Enter a number: 10
```

```
Pre-decrement = 9
```

```
Post-decrement = 9
```

```
Value after post-decrement = 8
```

```
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

8. Special Operators

XXXII. Important special operators in C include sizeof, comma.

// Program to demonstrate important special operators in C include sizeof, comma

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int a, b, sum;
```

```
    printf("Enter two numbers: ");
```

```
    scanf("%d %d", &a, &b);
```

```
    sum = (a, b, a + b);
```

```
    printf("Sum = %d\n", sum);
```

```
    printf("Size of a = %d bytes", sizeof(a));
```

```
    return 0;
```

```
}
```

Output:

```
Enter two numbers: 34 32
```

```
Sum = 66
```

```
Size of a = 4 bytes
```

```
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

XXIII. Write a program to find the size of int, float, char, and double using sizeof.

// Program to find the size of int, float, char, and double using sizeof

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    printf("Size of int = %d bytes\n", sizeof(int));
```

```
    printf("Size of float = %d bytes\n", sizeof(float));
```

```
    printf("Size of char = %d byte\n", sizeof(char));
```

```
    printf("Size of double = %d bytes", sizeof(double));
```

```
    return 0;
```

```
}
```

Output:

```
Size of int = 4 bytes
```

```
Size of float = 4 bytes
```

```
Size of char = 1 byte
```

```
Size of double = 8 bytes
```

```
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

XXIV. Write a program to use the comma operator and find the final value of a variable

1	2	3
int a = 10;a += 5;a *= 2;a -= 4;printf("%d", a);	int a = 5, b = 3;printf("%d", a & b);printf("%d", a b);printf("%d", a ^ b);	int a = 5;printf("%d\n", ++a);printf("%d\n", a++);printf("%d\n", a);
int a = 10;int b = a++;printf("%d %d", a, b);	int a = 10;int b = ++a;printf("%d %d", a, b);	int a = 10, b = 3;int c = a % b + a / b;printf("%d", c);
int a = 5,b=2;int c = a++ + ++b;printf("%d %d %d", a, b,c);	int a = 10,b=20;printf("%d", a > b && b > 5);	int a = 5;printf("%d", a << 1);printf("%d", a >> 1);

// 1. Assignment operators

```
#include <stdio.h>
```

```
int main()
{
    int a = 10;

    a += 5;
    a *= 2;
    a -= 4;
```

```
printf("%d", a);

return 0;
}
```

Output

```
26
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

// 2. Bitwise AND, OR and XOR

```
#include <stdio.h>
```

```
int main()
{
    int a = 5, b = 3;

    printf("%d\n", a & b);
    printf("%d\n", a | b);
    printf("%d", a ^ b);

    return 0;
}
```

Output

```
1
7
6
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

// 3. Pre-increment and Post-increment

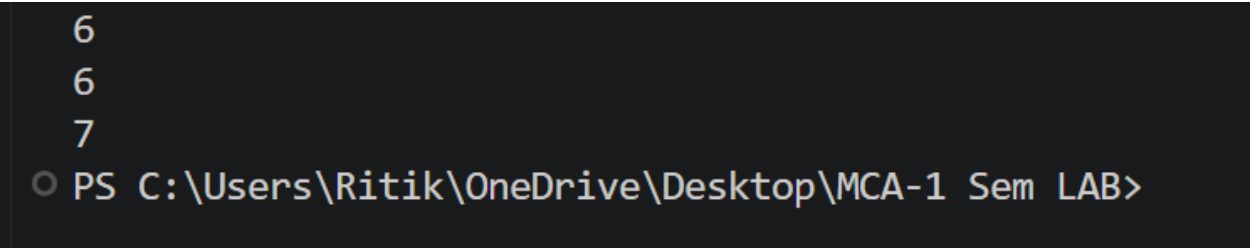
```
#include <stdio.h>
```

```
int main()
{
    int a = 5;

    printf("%d\n", ++a);
    printf("%d\n", a++);
    printf("%d", a);

    return 0;
}
```

Output



```
6
6
7
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

// 4. Post-increment using another variable

```
#include <stdio.h>
```

```
int main()
{
    int a = 10;
    int b = a++;

    printf("%d %d", a, b);

    return 0;
}
```

```
}
```

Output

```
11 10
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

// 5. Pre-increment using another variable

```
#include <stdio.h>
```

```
int main()
{
    int a = 10;
    int b = ++a;

    printf("%d %d", a, b);

    return 0;
}
```

Output:

```
11 11
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

// 6. Modulus and division operators

```
#include <stdio.h>
```

```
int main()
{
```

```
int a = 10, b = 3;
int c = a % b + a / b;

printf("%d", c);

return 0;
}
```

Output:

```
4
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

// 7. Post-increment and Pre-increment in expression

```
#include <stdio.h>

int main()
{
    int a = 5, b = 2;
    int c = a++ + ++b;

    printf("%d %d %d", a, b, c);

    return 0;
}
```

Output:

```
6 3 8
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

// 8. Logical AND operator

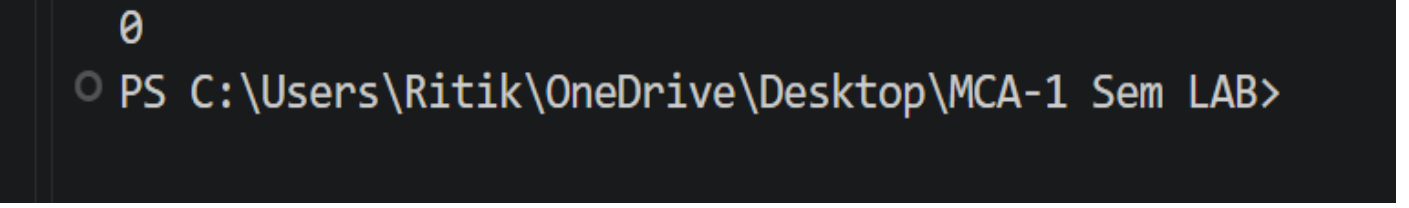
```
#include <stdio.h>
```

```
int main()
{
    int a = 10, b = 20;

    printf("%d", a > b && b > 5);

    return 0;
}
```

Output:



A screenshot of a terminal window with a dark background. The prompt is 'PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>'. The output of the program is '0'.

// 9. Left shift and Right shift operators

```
#include <stdio.h>
```

```
int main()
{
    int a = 5;

    printf("%d\n", a << 1);
    printf("%d", a >> 1);

    return 0;
}
```

Output:

10

2

PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>